A Mini Project Report on

SilentEcho: ASL to Text/Speech Converter

Submitted in partial fulfilment of the requirement for the

Bachelor of Technology (S.Y)

in

Computer Engineering

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Academic Year: 2025–2026

Approval Sheet

This is to certify that the Mini Project entitled

SilentEcho :- ASL to text/speech converter

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Declaration

We declare that this written submission for S.Y. Mini Project entitled **"SilentEcho :-ASL to text/speech converter"** represent our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the original sources. We also declared that we have adhere to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any ideas / data / fact / source in our submission. We understand that any violation of the above will cause for disciplinary action by institute and evoke penal action from the sources which have thus not been properly cited or from whom paper permission have not been taken when needed.

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ABSTRACT

Effective communication lies at the core of human interaction yet individuals with speech and hearing impairments often face significant challenges in expressing themselves clearly. This proposed system, named **SilentEcho**, introduces an intelligent real-time gesture-to-text and speech conversion framework designed to bridge this communication gap through the integration of computer vision and machine learning. The system leverages the **MediaPipe** framework for accurate hand landmark detection and utilizes a **Random Forest Classifier** to recognize gestures captured via a standard webcam. Each gesture, representing English alphabets (A–Z), is characterized by 21 distinct hand landmarks whose coordinate data are extracted, processed, and classified. The recognized gesture is then converted into text and synthesized into speech using the **pyttsx3** text-to-speech engine. Experimental results demonstrate that **SilentEcho** achieves high accuracy, smooth real-time performance, and reliability on standard consumer hardware without requiring specialized sensors. This work presents an affordable and accessible assistive technology that empowers individuals with speech and hearing disabilities. Future improvements aim to incorporate deep learning for continuous gesture recognition and context-aware communication to make interactions more natural and effective.

Keywords: gesture recognition, hand tracking, sign language, MediaPipe, Random Forest, text-to-speech.

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INTRODUCTION

Communication is a fundamental aspect of human existence—it allows individuals to share ideas, emotions, and experiences, thereby fostering understanding and collaboration. However, for individuals with speech and hearing impairments, effective communication can be an enduring challenge. The inability to express thoughts verbally often leads to social isolation, difficulty in education, and professional limitations. Sign language serves as a vital communication medium for the deaf and mute community, but a significant communication gap persists since most people are not familiar with it [1, 2].

Recent advancements in the fields of computer vision and machine learning have enabled the creation of systems that can recognize human gestures, movements, and hand signs in real time [3, 4]. This proposed system, named **SilentEcho**, leverages these technologies to convert American Sign Language (ASL) gestures into corresponding text and speech output, enabling seamless interaction between hearing-impaired individuals and non-signers. The system utilizes the **MediaPipe** framework for precise hand landmark detection and a **Random Forest Classifier** for gesture classification, ensuring efficient performance and high accuracy even on consumer-grade hardware [3].

Existing commercial applications such as **SignAll** [5], **HandTalk** [6], and **Microsoft's Kinect Translator** [7] have demonstrated the feasibility of real-time sign language interpretation, but they often require specialized sensors, large-scale setups, or internet connectivity. In contrast, **SilentEcho** is designed to operate offline, providing a cost-effective and accessible solution for personal and educational use. By transforming static gestures into text and speech, this proposed system

aims to bridge the communication barrier in everyday human interactions.

The integration of machine learning and assistive technology has expanded the potential of gesture-based communication systems. MediaPipe enables extraction of 21 hand landmarks per frame, which are used as feature inputs for the trained classifier. These features are then processed by the **Random Forest** model to predict the corresponding alphabet with remarkable precision. Furthermore, the use of the **pyttsx3** library ensures offline text-to-speech (TTS) conversion, eliminating the need for internet-based APIs. The result is a robust, lightweight, and fully autonomous assistive framework designed to enhance inclusivity through innovation.

Report Outline: The remainder of this report is structured as follows. Chapter 2 discusses related work and surveys existing systems, highlighting their methodologies and limitations. Chapter 3 describes the proposed system in detail, including its problem statement, objectives, and scope. Chapter 4 presents the design methodology, tools used, and implementation framework. Chapter 5 outlines the future scope and possible enhancements to the SilentEcho system. The report concludes with references and acknowledgements.

Chapter 2
Study of the system

1.1 Literature Survey

1.1.1 Survey on Hand Gesture Recognition from Visual Input

Linardakis *et al.* [1] present a comprehensive and up-to-date survey on hand gesture recognition across various input modalities such as RGB, RGB-D, and multi-view systems. The study categorizes research into key areas including gesture classification, 3D hand-pose estimation, multimodal fusion, and real-time on-device systems. It highlights challenges such as occlusion, dataset scarcity, and model generalization.

Tools/Technology Used: MediaPipe, RGB/RGB-D datasets, deeplearning models (CNNs, RNNs, Transformers).

Gap Identified: Lacks practical implementation for real-world, resourcelimited systems.

1.1.2 Study and Survey on Gesture Recognition Systems

Deshpande *et al.* [2] compare various gesture-recognition systems, exploring both vision-based and sensor-based modalities. The paper discusses applications in human–computer interaction, healthcare, robotics, and gaming. The authors analyze architectures and datasets, highlighting strengths and weaknesses of different methods and emphasizing the importance of integrating non-manual cues such as facial and body movements.

Tools/Technology Used: OpenCV, MediaPipe, IMU/Glove sensors, SVM, Random Forest, CNN, LSTM.

SVM, Random Forest, CNN, LSTM.

Gap Identified: Limited experimental validation and absence of real-time performance evaluation.

1.1.3 MediaPipe and CNNs for Real-Time ASL Recognition

Kumar *et al.* [3] propose a practical implementation of American Sign Language (ASL) alphabet recognition using a real-time pipeline combining MediaPipe and Convolutional Neural Networks (CNNs). The model achieves high accuracy on controlled datasets, demonstrating the efficiency of the **webcam → MediaPipe → CNN** workflow for low-cost, real-time applications.

Tools/Technology Used: MediaPipe for hand-landmark extraction, OpenCV for preprocessing, CNN classifier (TensorFlow/PyTorch).

Gap Identified: Tested only on static ASL alphabets; lacks validation on dynamic gestures and real-world variability.

1.1.4 Survey on Emotional Body Gesture Recognition

Noroozi *et al.* [4] focus on recognizing emotions through body gestures rather than facial or vocal cues. The paper reviews various datasets, kinematic features, and recognition methods, including classical and deep-learning-based techniques. It emphasizes challenges such as cultural variability, data scarcity, and lack of standardized benchmarks.

Tools/Technology Used: Pose/shape feature extractors, skeleton estimation, SVM, Random Forest, CNN, RNN models.

Gap Identified: Limited integration of deep neural networks and contextual emotion understanding.

1.2 Existing Systems

1.2.1 2.2.1 SignAll: Automatic Sign-Language Translation System

In the existing system, SignAll [5] is a commercial and academic platform designed to automatically translate sign language into text using a multi-camera computer-vision setup. It captures the signer's movements through multiple calibrated cameras and uses computer-vision algorithms to interpret gestures. The system aims to facilitate real-time communication between hearing-impaired and non-sign-language users, primarily in classrooms, business meetings, and customer-service applications.

Features:

- Multi-camera input setup for 3D motion capture.
- Uses computer vision and natural-language processing (NLP) for translation.
- Supports American Sign Language (ASL).

Drawbacks:

- Requires a controlled environment and complex camera calibration.
- High hardware and installation cost.
- Limited generalization to real-world, unconstrained settings.

1.2.2 HandTalk / Avatar Translators (Text/Audio Sign Translation)

In the existing system, HandTalk [6] and similar avatar-based translator systems focus on converting written or spoken language into animated

sign language using 3D avatars. These systems make digital content accessible to hearing-impaired individuals by translating text or speech into animated sign language in real time.

Features:

- Converts text or speech into 3D sign-language animations.
- Used for accessibility in websites, apps, and public-information systems.
- Integrates NLP and animation pipelines for gesture synthesis.

Drawbacks:

- Avatar movements lack natural human nuances such as facial expressions and non-manual cues.
- One-way communication: cannot translate gestures back to text or speech.
- Accuracy and emotional representation remain limited.

1.2.3 Microsoft Kinect Sign Language Translator (Research Prototype)

In the existing system, Microsoft Research developed the Kinect Sign Language Translator prototype [7], which uses RGB and depth data from Kinect sensors to recognize and translate sign language into spoken or written text. The system maps body and hand positions to predefined sign gestures using machine-learning algorithms.

Features:

- Uses Kinect's RGB and depth sensing for skeletal tracking.
- Recognizes sign-language gestures in constrained environments.
- Converts recognized gestures into text or speech output.

Drawbacks:

- Dependent on Kinect hardware, which is discontinued.
- Performs best under controlled lighting and distance.
- Struggles with complex sentences and non-manual expressions.

1.3 Comparison of Existing Systems

Table 1.1: Comparison of Existing Sign-Language Translation Systems

System Name	Input Type	Core Technology Used	Output Type	Advantages	Limitations
SignAll [5]	Multi-camera (3D capture)	Computer Vision, NLP	Text/Speech	High accuracy; supports full ASL translation; professional-grade performance	Expensive setup; requires multiple cameras and controlled environment
HandTalk / Avatar Translators [6]	Text or Audio input	NLP, 3D Animation, Avatars	Animated Sign Language	Web and mobile accessible; interactive visualization	One-way translation only; lacks facial expression modeling
Microsoft Kinect Sign Translator [7]	RGB + Depth (Kinect Sensor)	Skeleton Tracking, Machine Learning	Text/Speech	Real-time detection; depth-assisted gesture tracking	Requires proprietary hardware; sensitive to lighting and range
SilentEcho (Proposed System)	Single Webcam	MediaPipe, Random Forest Classifier, pyttsx3	Text/Speech	Low-cost, real-time, lightweight; accessible using standard webcam; runs offline	Limited gesture vocabulary; no continuous sentence formation

From this comparison, it is evident that most existing systems either depend on costly or specialized hardware (**Kinect**, multi-camera rigs) or offer one-way avatar-based translation. **SilentEcho** introduces an affordable, webcam-based alternative that performs real-time gesture recognition and text-to-speech conversion locally, making it accessible and efficient for everyday use.

Chapter 2

Proposed System

2.1 Problem Statement

Communication is one of the most essential aspects of human existence—it forms the foundation of relationships, learning, and collaboration. Yet, for millions of individuals with speech and hearing impairments, expressing ideas and emotions remains a constant challenge. Despite the availability of sign language, the lack of universal understanding among the general population continues to create barriers in daily communication, education, and employment.

This proposed system, named **SilentEcho**, is a step toward bridging this communication divide by enabling gesture-based interaction that converts sign gestures into both text and speech in real time. The goal is to create a system that is not only functional but also empathetic, making technology a medium of inclusion and understanding [1–3].

Key Challenges:

- **Limited Accessibility:** Sign language is effective but not widely known, restricting meaningful interaction between hearing- and speech-impaired individuals and the general public.
- **Hardware Dependency:** In the existing systems, assistive devices often rely on sensor-based gloves or depth cameras, which are expensive, non-portable, and unsuitable for daily use [5, 7].
- **Intrusive Design:** Many existing solutions require users to wear

external devices, making them uncomfortable and impractical for continuous interaction.

- **Lack of Real-Time Responsiveness:** Some deep-learning-based systems depend on cloud processing, resulting in higher latency and reduced real-world usability.

Need for the System:

- **Cost-Effective and Accessible:** Should eliminate the need for specialized hardware while remaining affordable for individuals and institutions.
- **Real-Time and Efficient:** Must process gestures instantly to enable natural, uninterrupted communication.
- **User-Friendly and Non-Intrusive:** Should function with commonly available devices like webcams without requiring wearable sensors.
- **Scalable and Adaptive:** Should be deployable across multiple environments such as classrooms, workplaces, and healthcare centers.
- **Inclusive and Empowering:** Should promote independence and confidence among individuals with speech and hearing impairments by enabling seamless communication.

2.2 Scope

The **SilentEcho** project focuses on designing and implementing a real-time, accessible, and efficient communication system for individuals with speech and hearing impairments. The system recognizes commonly used hand gestures through computer vision techniques and con-

verts them into text and audible speech output, enabling meaningful interaction with people who do not understand sign language.

The project utilizes the **MediaPipe** framework for accurate hand-landmark detection and a **Random Forest Classifier** for gesture recognition, ensuring high accuracy and lightweight performance. It operates efficiently on standard consumer hardware, eliminating the need for expensive sensors or high-end devices [3, 7].

The scope of SilentEcho extends to:

- Assisting differently-abled individuals in daily communication and social interaction.
- Enabling usage in educational, healthcare, and workplace environments where interpreters are not always available.
- Providing a foundation for future research in gesture-based communication, including full sign-language translation and emotion-based gesture recognition [4].
- Offering a scalable framework that can be enhanced using deep learning models, multi-hand recognition, and multi-language speech synthesis for broader accessibility.

In essence, this proposed system demonstrates how technology and empathy can combine to create inclusive solutions, transforming silent gestures into a universal language of connection and understanding.

2.3 Hardware Requirements

The hardware components required for the development and execution of **SilentEcho** are minimal and easily accessible. The system is designed to run efficiently on standard computing devices without the need for specialized or expensive equipment.

Hardware Component	**Specification / Description**
Laptop / Computer	Intel Core i5 processor (or equivalent), minimum 4 GB RAM, 5 GB storage
Webcam	Integrated or external webcam for real-time gesture capture
Microphone (optional)	For audio testing and speech output validation
Speaker / Headphones	For audio output from the text-to-speech module
Power Supply	Standard 230V AC, 50Hz
Internet Connection	Required only for initial setup, library installation, and dataset access

Table 2.1: Hardware Requirements for SilentEcho

The system operates using only a webcam and a standard computing device, making it a low-cost and portable solution suitable for classrooms, workplaces, and healthcare facilities.

Chapter 3

Design of the System

3.1 Approach

This proposed system, named **SilentEcho**, focuses on real-time translation of hand gestures into text and speech using computer vision and machine learning. The system integrates multiple modules that work together seamlessly to enable natural and intuitive gesture-based communication. These modules include gesture capture, hand-landmark extraction, gesture classification, text generation, and speech synthesis.

The primary goal of this approach is to develop an assistive tool that functions efficiently in real time, operates on standard consumer hardware, and requires no specialized sensors. By leveraging the **MediaPipe** framework for precise landmark detection and a **Random Forest Classifier** for gesture prediction, the system ensures both accuracy and efficiency while maintaining simplicity and accessibility [1,3,7]. This modular design ensures that every component—from gesture input to audio output—contributes to a robust and reliable real-time communication experience.

3.1.1 Flow Chart

The flowchart shown in Figure 3.1 illustrates the end-to-end working of the **SilentEcho** system. It depicts the process beginning with webcam input, followed by frame processing, hand-landmark detection, gesture recognition, and conversion into corresponding text and speech in real

time [2, 3].

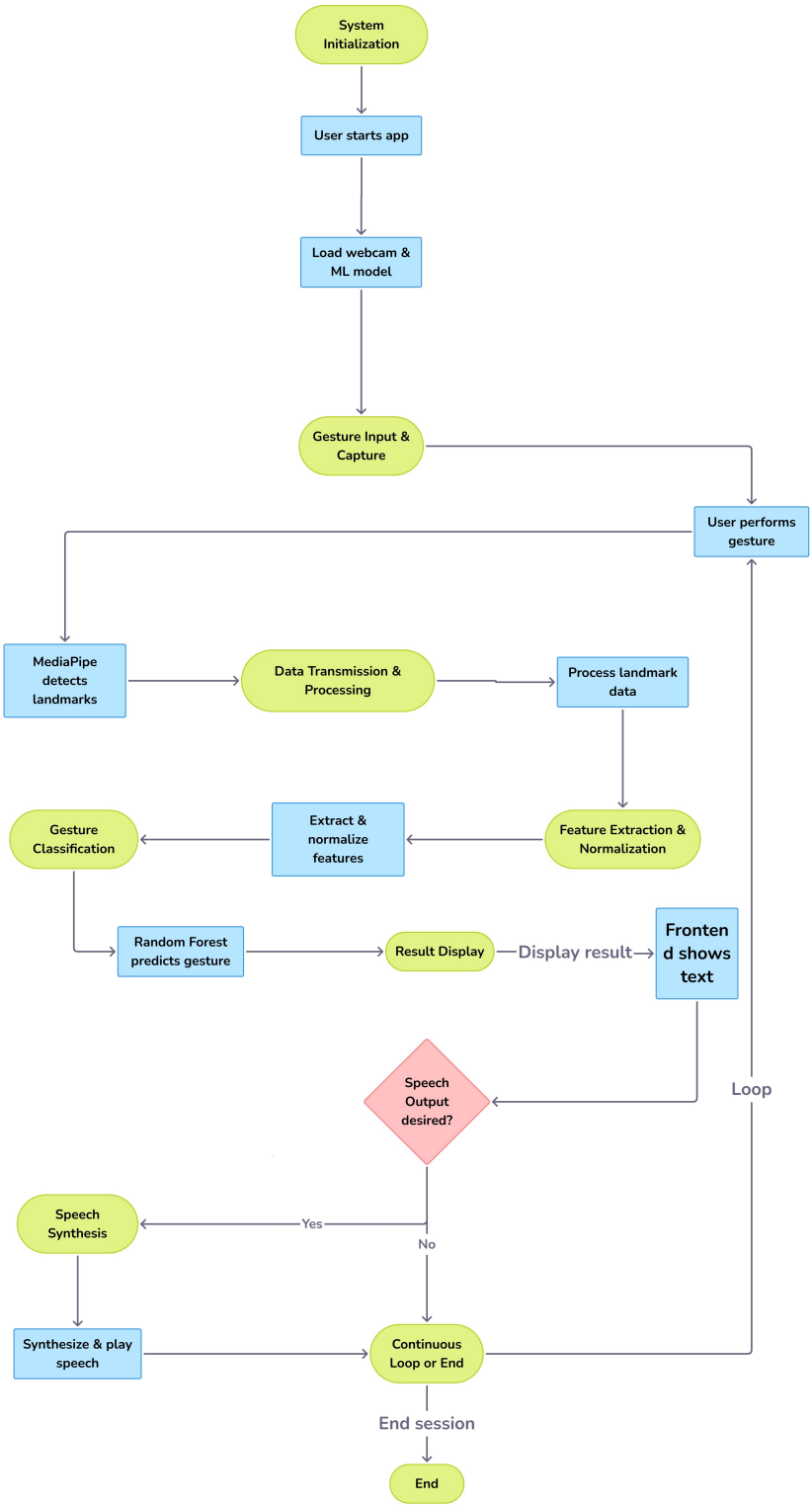

Figure 3.1: Flowchart illustrating the overall workflow of the **SilentEcho** system.

3.1.2 ASL Alphabet Reference

The system follows the **American Sign Language (ASL)** alphabet standard for gesture recognition, as shown in Figure 3.2. Each English alphabet (A–Z) corresponds to a unique static hand gesture that serves as the foundational dataset for training and testing.

Using the **MediaPipe** framework, 21 hand-landmark coordinates are extracted from each ASL gesture. These coordinates form the feature set for the **Random Forest Classifier**, allowing accurate classification of gestures with minimal computational complexity. This ASL-based dataset ensures a structured and standardized mapping between hand poses and English alphabets, enabling the **SilentEcho** system to translate gestures into meaningful text and speech outputs [3, 5].

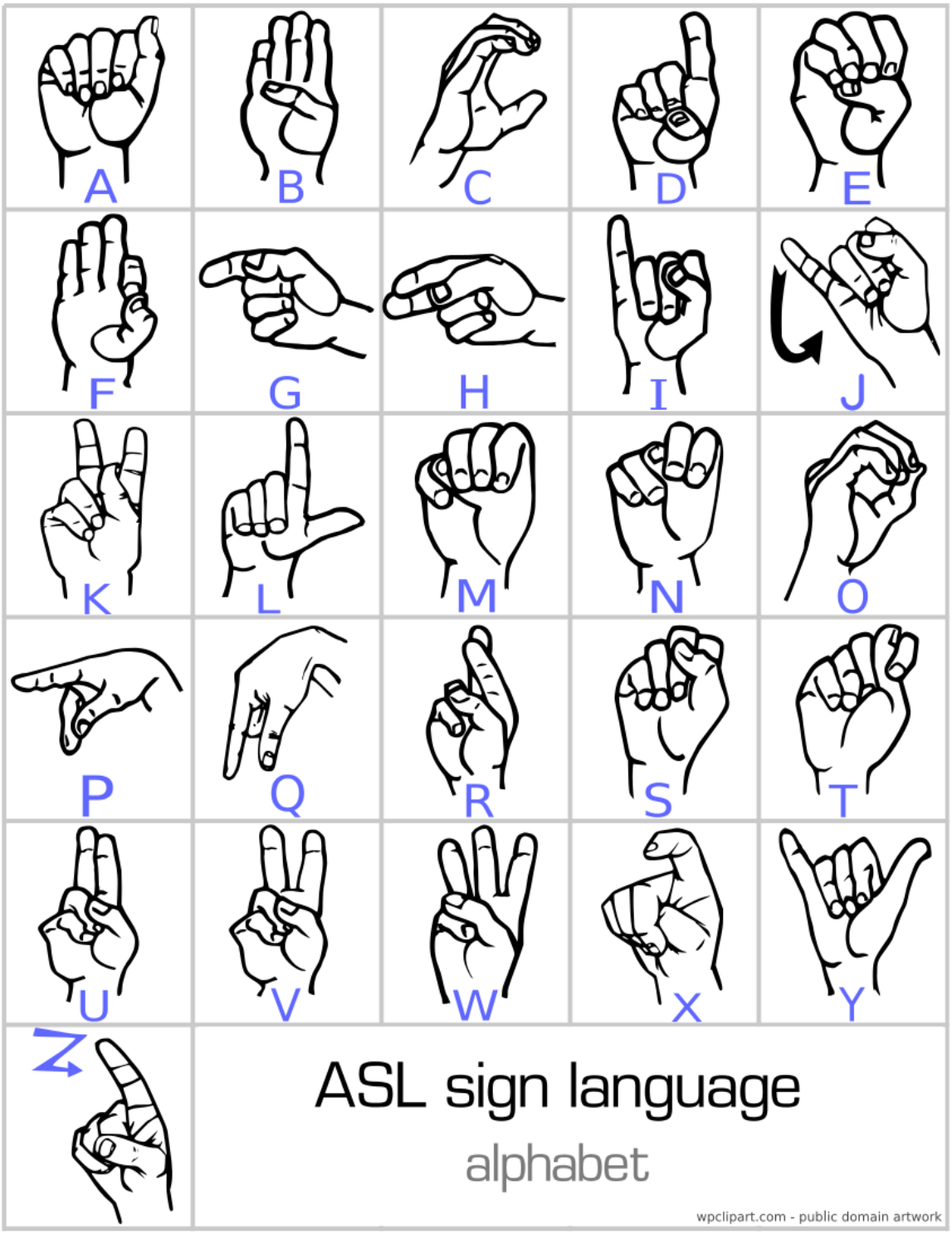

Figure 3.2: American Sign Language (ASL) alphabet reference used for gesture recognition in the **SilentEcho** system.

3.2 Technology Stack

The **SilentEcho** system is developed using a combination of software tools and frameworks that enable efficient computer-vision processing, machine learning, and real-time text-to-speech conversion. The technology stack has been carefully chosen to ensure optimal performance, scalability, and platform independence.

3.2.1 Tools Required

This section outlines the tools and technologies used in both the **front end** and **back end** of the system.

Front End

The front end of **SilentEcho** provides a user-friendly and responsive interface for real-time interaction with the gesture recognition system. It enables users to visualize detected gestures and receive immediate text and speech output.

Technologies Used:

- **HTML5:** Structures the web interface, including live video display, control buttons, and text output panels.
- **CSS3:** Ensures responsive and accessible styling across devices.
- **JavaScript (Fetch API):** Manages asynchronous communication between frontend and backend for gesture predictions and speech updates.
- **Flask Templates:** Renders `index.html` and integrates live video streams from the backend.
- **OpenCV (Frontend Display):** Streams frames from the webcam to the browser for real-time visualization.

- **Browser SpeechSynthesis API:** Converts recognized text to speech directly within the web interface.

Key Features:

- Real-time webcam video streaming.
- Instant gesture prediction and text display.
- Interactive buttons for mode switching and speech control.
- ASL reference chart pop-up for guidance.
- Fully responsive cross-platform interface.

Back End

The backend handles all system logic, including gesture recognition, data processing, and text-to-speech synthesis. It ensures low-latency communication and efficient machine-learning inference.

Tools and Libraries Used (Version Specific):

- **Python 3.8.10:** Core programming language for backend development.
- **Flask 2.0.3:** Micro web framework for routing and API management.
- **MediaPipe 0.10.0:** Framework for real-time hand-landmark detection (21 landmarks per frame) [3].
- **OpenCV 4.5.5:** Used for video frame acquisition, preprocessing, and image enhancement.
- **scikit-learn 1.2.0:** Implements the Random Forest Classifier for gesture recognition.
- **pyttsx3 2.90:** Provides offline text-to-speech conversion.

Backend Responsibilities:

- Capture and preprocess webcam input using OpenCV.
- Extract hand landmarks via MediaPipe.
- Predict gestures with the trained Random Forest model.
- Convert recognized text to audio using `pyttsx3`.
- Communicate predictions to the frontend through Flask APIs.

The integration of these tools ensures that **SilentEcho** achieves accurate gesture recognition and efficient speech synthesis on standard hardware—making it an affordable and accessible assistive communication tool [2, 3, 7].

3.3 System Architecture

3.3.1 Block Diagram

The **System Architecture** of **SilentEcho** defines the logical flow of data between its functional components. It comprises five main modules: Gesture Acquisition, Feature Extraction, Classification, Text Conversion, and Speech Synthesis. Each module plays a critical role in ensuring precise real-time translation from gestures to audio output [3, 5, 7].

The block diagram shown in Figure 3.3 visually represents this architecture. The **Input Module** captures hand gestures using a standard webcam. These frames are processed by the **MediaPipe Framework** to extract 21 hand-landmark coordinates. The features are passed into a trained **Random Forest Classifier**, which predicts the corresponding alphabet. The recognized text is then displayed on-screen and converted to speech via the **pyttsx3** engine.

This modular and layered architecture allows independent testing and debugging of each subsystem, maintaining synchronization across modules and achieving efficient, real-time performance.

Block Diagram: The following figure illustrates the complete functional workflow of the **SilentEcho** system.

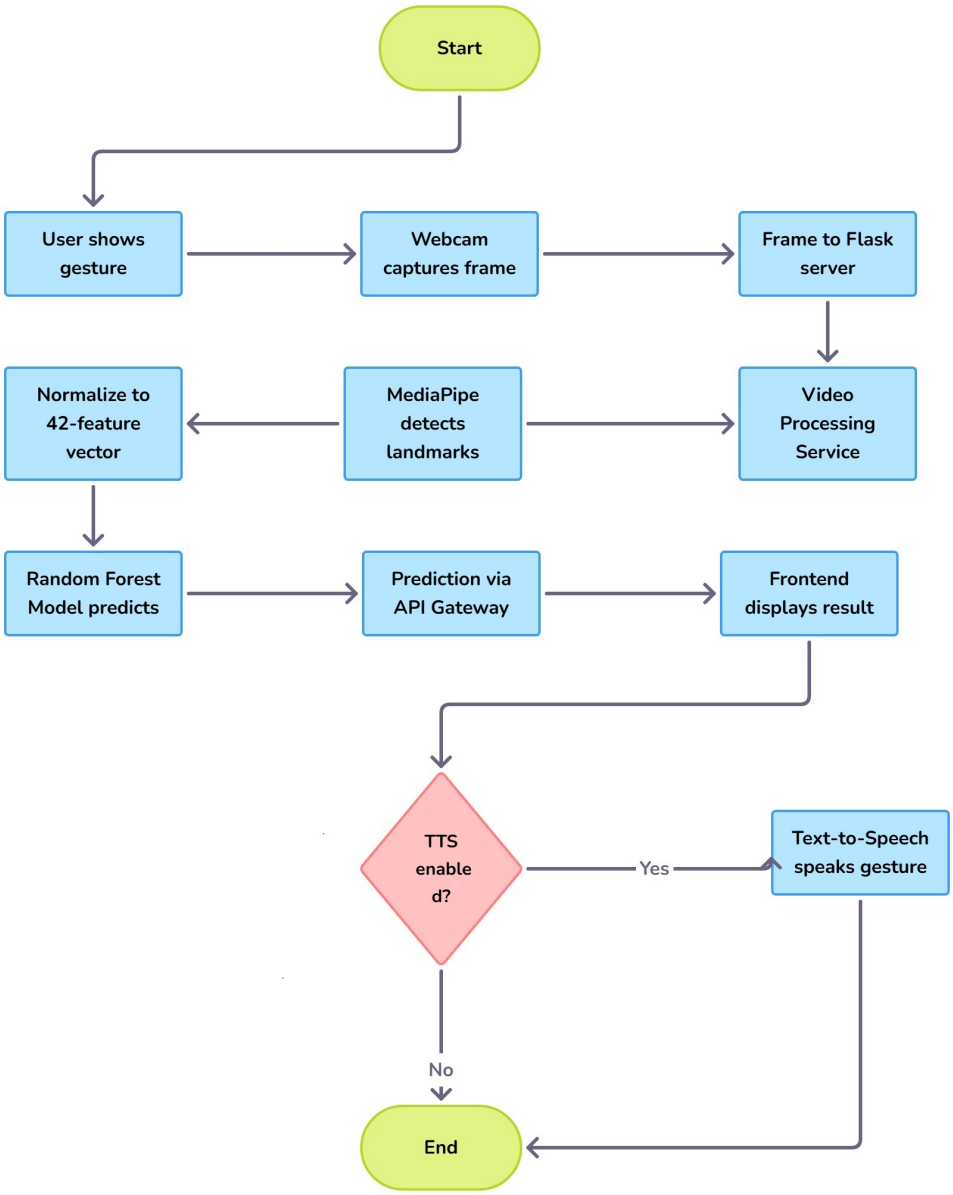

Figure 3.3: System architecture and block diagram of the **SilentEcho** system.

Acknowledgement

Success of a project like this involving high technical expertise, patience, and massive support of guides, is possible when team members work to-gather. We take this opportunity to express our gratitude to those who have been instrumental in the successful completion of this project. We would like to appreciate the constant interest and support of our mentor guide **Mrs.Chetana Badgujar** in our project and aiding us in developing a flair for the field of Your Domain. We would always cherish the journey of transforming the idea of our project into reality. We would like to show our appreciation to **Mrs.Chetana Badgujar** for his/her tremendous support and help, without whom this project would have reached nowhere. We would also like to thank our project coordinator **Mrs.Chetana Badgujar** for providing us with regular inputs about documentation and project timeline. A big thanks to our **HOD Dr. M. Kiruthika** for all the encouragement given to our team. We would also like to thank our principal, **Dr. S. M. Khot**, and our college, **Fr.C.Rodrigues Institute of Technology, Vashi**, for giving us the opportunity and the environment to learn and grow.

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Appendix A: Timeline Chart

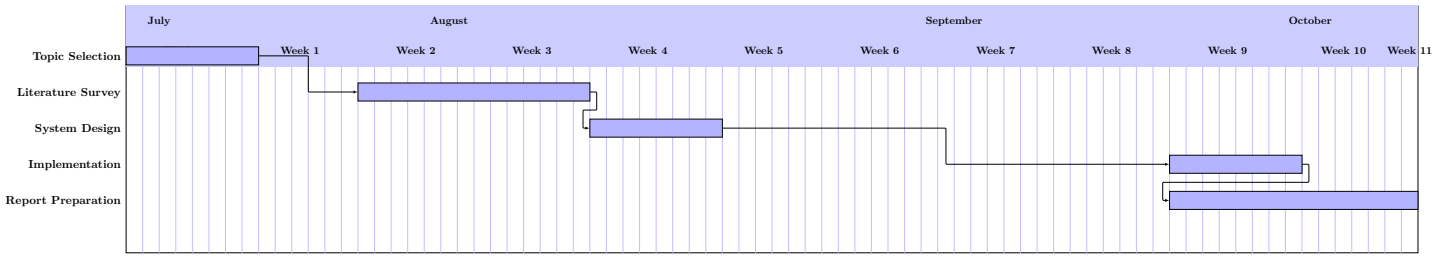

Figure A.1: *Project Timeline for SilentEcho Development – illustrating all major phases from topic selection to report preparation.*